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Automated Integrated Workflow for Enhanced Reservoir Characterization and History Matching Using Machine Learning and Ember in Complex Reservoir Systems

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Abstract

The increasing complexity of modern reservoirs has emphasized the need for advanced, integrated, and automated workflows that efficiently manage geological and dynamic uncertainties. This study presents a comprehensive methodology that couples static and dynamic reservoir modeling within a single simulation environment, using machine learning and Python scripting to enhance automation, accuracy, and speed. The proposed workflow includes automated static property modeling using the EMBER algorithm and dynamic model calibration through a machine learning-assisted history matching process. Over 20 equilibrium regions, commingled production, and two reservoir fluids were modeled under limited data conditions, using probabilistic scenarios, experimental design, and geostatistical inputs to reduce uncertainty and improve model fidelity.

The results demonstrated that the integrated workflow not only accelerated the generation of simulation ensembles but also significantly improved model predictability and calibration quality. SHAP-based sensitivity analysis enabled the identification of high-impact variables, and visual tools such as tornado and worm plots enhanced interpretability. Compared to traditional approaches, this method reduced computational time and human error, resulting in a more reliable basis for field development planning. The novelty of this approach lies in the seamless integration of machine learning, probabilistic modeling, and workflow automation in a unified platform, offering a replicable strategy for efficient and scalable reservoir management in highly uncertain environments.

Introduction

The volatility of oil prices in recent years has driven the oil and gas industry to seek greater efficiency in all aspects of hydrocarbon exploration, modeling, and production. Reservoir characterization and dynamic simulations continue to be essential tools for understanding fluid flow behavior and predicting reservoir performance. Because conventional, easily accessible oil reserves have largely been exploited, the industry is focusing more on complex reservoirs. These reservoirs include deepwater fields, low-permeability formations, naturally fractured systems, thin-pay reservoirs, and accumulations with challenging flow properties. The increasing complexity of both subsurface and surface production systems has made uncertainty quantification a necessary step in reservoir modeling. (Kumar et al. 2017).

To address these challenges, the industry has shifted from deterministic approaches, where only a limited set of predefined geological models is used, to probabilistic workflows that incorporate a wider range of uncertainties. In probabilistic modeling, multiple input parameters are varied systematically, generating a distribution of possible reservoir performance results. These workflows have become standard for history matching, production forecasting under uncertainty, model updating with new data, and field development optimization under uncertainty. (Kumar et al. 2017).

A key element in reservoir characterization and history matching is the integration between static and dynamic modeling. Traditional two-step workflows for uncertainty assessment and probabilistic history matching often insufficient to address the complexities of geological uncertainties. For example, combining geological uncertain parameters into a few earth models can obscure the direct impact of these uncertainties on forecasts, leading to artificial correlations that complicate history matching. (Kumar et al. 2017).Toalleviate these limitations, big loop workflows were introduced, encompassing the adjustment of input parameters in the static or geological model and performing uncertainty analysis to improve history matching and optimize field development through prediction runs. (Chong et al. 2015). Due to the large number of uncertain parameters involved, evaluating all possible combinations is impractical. Instead, experimental design, also referred to as design of experiments, is used to efficiently explore the uncertainty space. ED is a statistical method that enhances process understanding by performing a limited set of carefully selected experiments (Goodwin et al. 2021). Traditionally, history matching of a reservoir model can be a tedious and resource-demanding process, but it is essential for understanding reservoir behavior. Conventional methods involve varying key uncertainty parameters, making sensitivity runs, and comparing results with observed data. However, these methods are often not fully automated and require additional utility applications to manage and process input data and model results. (Chong et al. 2015). Machine learning techniques have further enhanced this process by providing data-driven predictions and decisions, reducing the number of simulation runs needed to find a satisfactory solution, and significantly decreasing computational time. (Victoria et al. 2023). An integrated reservoir study using structured and novel machine learning and data analytics approaches has been conducted for history matching a giant mature multilayered oil field in the Mahakam Delta of Indonesia. This study highlights the unique reservoir modeling challenges, and the innovativeness of the data science methods employed. One of the most important elements in reservoir characterization and history matching is the integration between static and dynamic modeling. With numerous layers and many uncertain parameters, from dynamic uncertainties such as hydrocarbon contacts and communication between regions to static properties such as porosity and water saturation, hundreds of possible scenarios are created during history matching. Using Python scripting embedded in the reservoir simulator and extensive computational resources, these uncertainties can be handled quickly, and each ensemble can be analyzed easily with data analytics and machine learning-based proxy approaches. (Suwito et al. 2022).

In the other hand, geostatistical modeling techniques have long been employed as a standard approach for estimating petrophysical properties and conducting reservoir characterization. However, these methods rely on a set of assumptions that must be met before modeling to avoid potential risks. (Ferreira 2023). Machine learning algorithms such as EMBER @slb combine geostatistics with quantile Random Rorests algorithm to create a modeling methodology capable of overcoming these challenges. By incorporating multiple secondary variables and adopting a nonstationary spatial modeling approach, EMBER offers a promising solution for more precise and efficient reservoir modeling. (Ferreira 2023).

This project's goal was to model a complex field with around 20 equilibrium regions, two main reservoir fluids, commingled production, and high uncertainty due to limited data availability. Automated workflows were implemented from static to dynamic modeling, using different machine learning approaches to achieve reliable results in less time. For example, machine learning was used for property modeling and history matching. The workflow allows changes with a single click, from static variations to new simulation runs. This integrated approach not only achieves a robust depiction of subsurface uncertainty but also provides a higher degree of predictability in forecast scenarios, which is essential for quantifying the risk associated with field development plans.

The objective of this study is to develop an automated, repeatable, and integrated workflow that couples static and dynamic reservoir modeling to enhance uncertainty quantification and improve forecasting accuracy. Traditional approaches to history matching and uncertainty assessment often rely on sequential and computationally expensive methods that fail to fully capture the interaction between geological and dynamic uncertainties. This research proposes a novel methodology in which static and dynamic variations are jointly incorporated within a single simulation workflow, generating an ensemble of reservoir realizations that account for subsurface uncertainty in a more systematic manner. By leveraging Python scripting and machine learning techniques, the workflow automates key processes such as property population and history matching, enabling rapid generation of stochastic reservoir simulation models. The integration of these techniques not only accelerates model calibration but also enhances predictive capabilities, allowing for more informed decision-making in field development and reservoir management.

Objectives and Scope

The objective of this study is to develop an automated, repeatable, and integrated workflow that couples static and dynamic reservoir modeling to enhance uncertainty quantification and improve forecasting accuracy. Traditional approaches to history matching and uncertainty assessment often rely on sequential and computationally expensive methods that fail to fully capture the interaction between geological and dynamic uncertainties. This research proposes a novel methodology in which static and dynamic variations are jointly incorporated within a single simulation workflow, generating an ensemble of reservoir realizations that account for subsurface uncertainty in a more systematic manner. By leveraging Python scripting and machine learning techniques, the workflow automates key processes such as property population and history matching, enabling rapid generation of stochastic reservoir simulation models. The integration of these techniques not only accelerates model calibration but also enhances predictive capabilities, allowing for more informed decision-making in field development and reservoir management.

Methodology

The methodology for this study involves several key steps to ensure a comprehensive evaluation of CCUS methodologies and EOR techniques in the Colombian field. The process is divided into the following stages:

This study developed a comprehensive and automated workflow to enhance the efficiency and accuracy of reservoir modelling and history matching. By integrating static and dynamic parameters, the workflow addresses all observed uncertainties impacting history matching, reducing human error and accelerating repetitive tasks. Machine learning techniques, including the EMBER algorithm, were employed to enhance property modelling and reservoir characterization, providing precise estimations and overcoming traditional geostatistical challenges. Additionally, machine learning was used in history matching to improve data-driven predictions and decisions, reducing the number of simulation runs needed and decreasing computational time.

Integrated Automated Workflows

One of the primary objectives of this project was to develop automated workflows focused on history matching. A comprehensive workflow was developed, integrating both static and dynamic parameters, including all parameters with observed uncertainties that directly impact history matching. Additionally, the workflow proved highly useful after uncertainty runs because it allowed for the recreation of the workflow with assigned variable values, facilitating the creation of new runs with only specific changes.

The workflow developed in this project uses an E&P software platform as an integrator tool and consists of a main workflow, which includes three secondary workflows representing sand probability, property modeling calculators, and aquifers. The main workflow, in addition to incorporating the secondary workflows, encompasses all numerical expressions where values are loaded to be changed either randomly (based on experimental design sampling) or manually during simulation runs, especially those involving uncertainty. The workflow also includes processes related to the creation of contacts, fluid models, and rock-fluid models. This workflow encompasses the creation of workflows through process logic and functions within the E&P software platform, enabling iterations of processes using different processes included in the software. Fig. 1 shows sections of some code lines of the workflow; on the right-hand side, the main workflow is shown, which indicates running the three different workflows on the left.

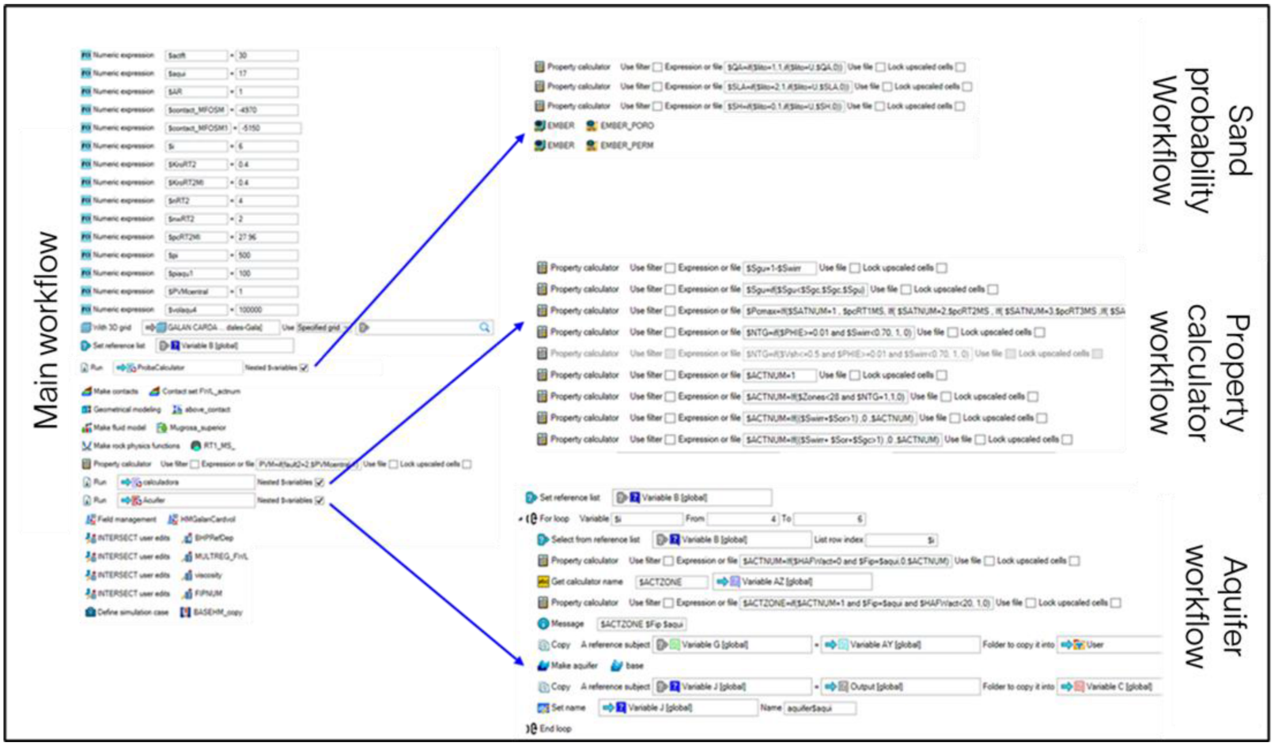


Figure 1—(a) Section of the main workflow and (b) secondary workflow.

Three secondary workflows comprise the solution. The first workflow is based on the sand and/or shale probability, which is used as an input for property modeling. The second workflow involves property calculation, encompassing rock properties and water saturation, which depend on petrophysical changes and the free water level respectively. The third workflow relates to the creation of aquifers for every region where the presence of an aquifer is possible.

The static model was created based on probability maps of lithotype and shale occurrences. This approach aims to incorporate these maps into property modeling using the EMBER machine learning algorithm. When the probability maps change, a repopulation occurs, which increases or decreases the likelihood of having high values of porosity and the corresponding calculation of permeability. This process results in a repopulated static model for each possible uncertainty scenario. These parameters allow for establishing disconnection and connection in certain zones or regions where conventional population can be optimistic or pessimistic in the dispersion of properties, especially where there is no control of values by wells. This inclusion of the probability parameter within the known field uncertainty ensures a more accurate representation.

Fig. 2(a) shows the workflow and the EMBER process within the flow. The figure shows the selection of regions to vary as well as the probability value, both indicated with the $ symbol to denotevariables. Fig. 2(b) shows two realizations of porosity in a layer where the probability of shale occurrence changed.

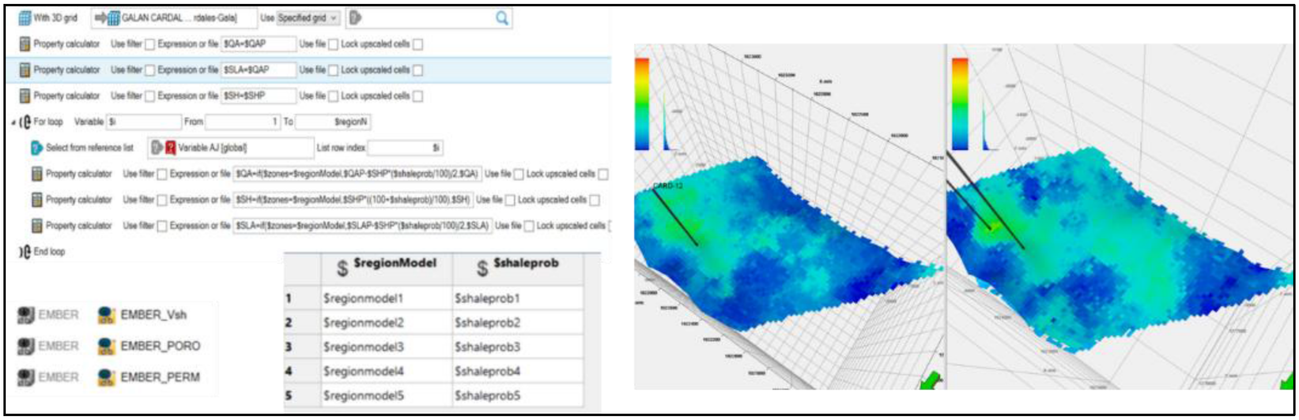


Figure 2—Workflow and EMBER process with variable selections and porosity realizations.

The second workflow involves the calculation of grid properties. Once the model has been repopulated, along with changes in the oil/water contact, it is necessary to recalculate properties such as rock type, quality, pore throat sizes, height to contact, capillary pressures, and saturations (according to known petrophysical equations). This workflow allows for recalculating all these properties based on the variable changes made in each run. Additionally, the workflow accounts for zones considered inactive due to petrophysical cutoffs.

The third workflow involves the creation of aquifers that are connected to each of the different zones where there is oil/water contact. When the contact changes, it is necessary to recalculate the zones where the aquifers will be connected, not only at the contact level but also considering active cells and the volume of water below the contacted water zone. Each aquifer within the 20 possible regions might or might not exist, along with their different combinations. A prior production analysis was conducted to narrow the range, but seven possible acquirers with different combinations and variations in their uncertain properties, such as water volume and productivity, were evaluated. Fig. 3 shows some lines of code from the workflow with the variable table selecting the zone where the aquifer might exist, the volume, and the productivity index for that zone. Fig. 3(a) shows one of the realizations, which in this case only presents two connected aquifers at different levels and regions.

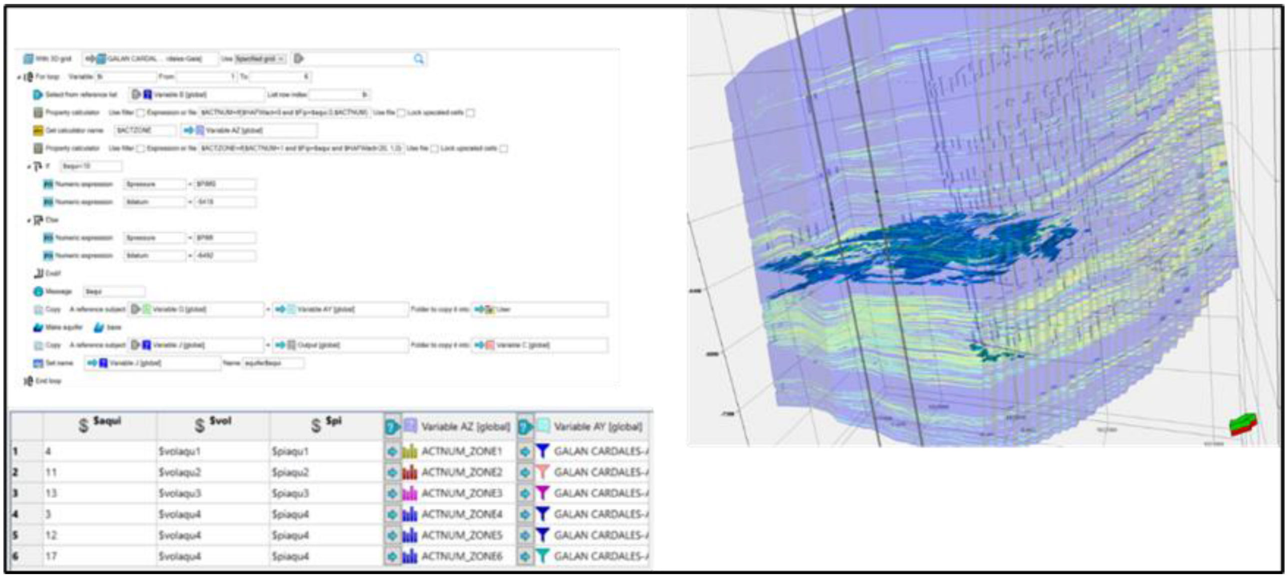


Figure 3—Aquifer workflow and realizations.

Machine Learning for Property Modeling (EMBER)

The Embedded Model Estimator (EMBER) is a petrophysical modeling algorithm designed to manage complex and heterogeneous reservoirs. This algorithm uses a combination of secondary variables, such as facies models and seismic attributes, to accurately determine effective porosity and permeability. In this project, EMBER was applied to populate porosity and permeability based on clay volume and the probability of shale and different sand occurrences. Each uncertainty scenario resulted in the creation of a different static grid

Machine Learning for History Matching

This project is characterized by its high heterogeneity and lenticular nature, presenting several uncertainties that needed to be addressed for effective history matching. The defined uncertainties included depth of contacts, relative permeability parameters, variables related to numerical aquifers, maximum capillary pressure, porous volume multiplier at the grid edges (to account for connections with neighboring areas), and clay probability to represent the level of communication between sands.

The history-matching process was assisted by machine learning, employing an iterative and optimized approach to identify the best combinations of uncertainty parameters in the simulation model. Initially, 40 relevant uncertainty variables were selected from all of those considered, taking into account time constraints and the requirement for simulation runs to train the model.

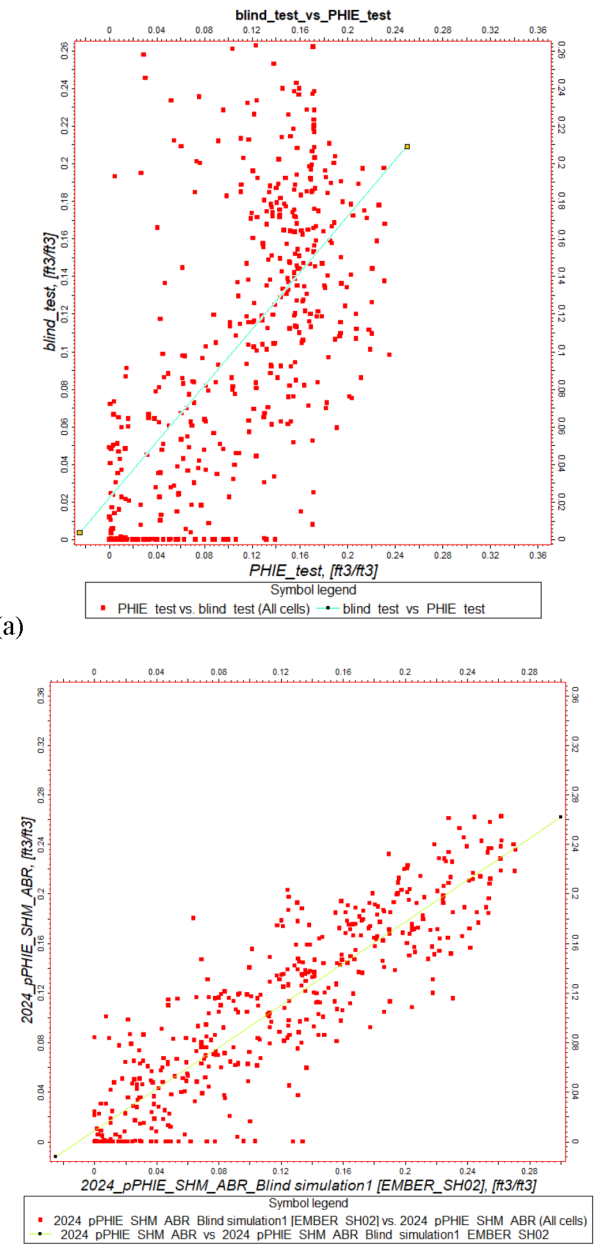
To define the minimum and maximum values for Monte Carlo simulations, initial runs with extreme values were conducted. These extreme values qualitatively determined whether they could reproduce curves that encompassed both above and below the observed data, ensuring adequate training in the machine learning process. Based on the extreme values obtained using a previous Plackett-Burman experiment design, 300 cases were performed to train the model, define important variables, and carry out uncertainty runs. This process was repeated three times until satisfactory results were achieved. Although runs with extreme values were conducted, these runs only showed the maximum and minimum curves that can be found in uncertainty runs. However, Monte Carlo runs were necessary to ensure random value selection for training.

An objective function was created to define a target during the machine learning training process. This function considers cumulative oil production, water cut, zonal pressures, and flowing bottomhole pressures. Because cumulative oil production (Np) is a time-accumulative vector, this objective function gave greater weight to wells with higher production, especially in the later years of the reservoir's life, ensuring the model's predictability, given that the wells with high accumulations had greater weight when calculating this objective function.

These results were used to train a machine learning model based on the Random Forest algorithm, which generated 10,000 simulations using random sampling and SHAP (Shapley Additive Explanations) values analysis. This analysis identified the top 100 runs based on their impact on the objective function.

The best-selected runs were analyzed to define ranges for key variables and retrain the model, iterating the process. Finally, the top 50 optimized results were obtained, focusing on maximizing the objective function.

Results



(a)

(b)

Figure 4—Comparison of (a) conventional petrophysical model and (b) EMBER model in blind test.

A blind test was performed in one of the wells excluded from the training dataset to validate the porosity population. Fig. 4 compares the porosity from well logs with the populated porosity values using two different approaches: (a) a conventional petrophysical modeling workflow (Petro Model) and (b) the machine learning algorithm EMBER.

The scatter plots show the dispersion of points around the 45° line, which represents a perfect match between measured and populated porosity. The conventional petrophysical model exhibits higher scatter, indicating larger deviations from the true log porosity. In contrast, the EMBER machine learning model provides a closer alignment with the reference line, reducing the prediction error and improving the representativeness of porosity distribution in blind wells.

This result demonstrates that ML-based population not only captures the underlying geological variability but also increases predictive accuracy when applied to wells outside the training set, highlighting the robustness of the workflow under uncertainty.

To determine which variables had the greatest impact on the objective function results and how their increase or decrease affected the match, several SHAP graphs were generated. The analysis of these graphs allowed for the identification and exclusion of less relevant variables in uncertainty runs; thus, improving the model training process.

The diagram shown in Fig. 5 illustrates tornado plots and worm plots, which are instrumental in understanding the significance of various variables and their impact on the objective function. These visual tools enable engineers to refine their decision-making process for subsequent iterations within the history matching workflow, ensuring a more accurate and efficient match.

The images in Fig. 6b are histograms of the posterior distribution of the best cases after 10,000 runs of the trained machine learning model. These histograms illustrate the optimal combinations of variables focused on minimizing error.

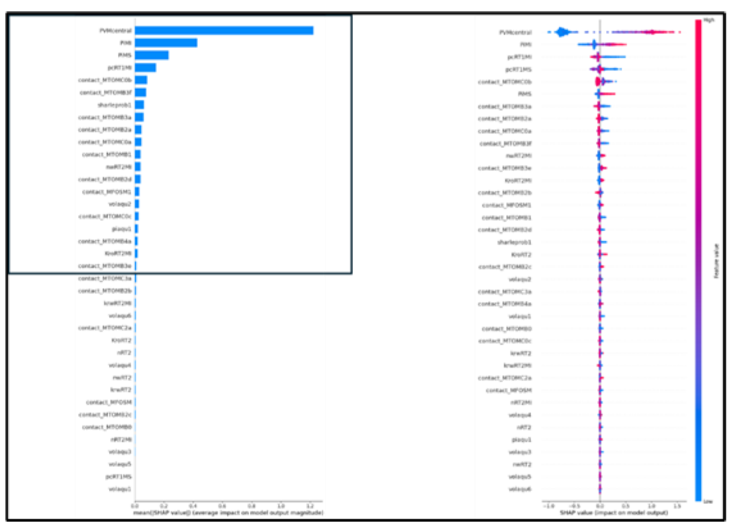


Figure 5—SHAP analysis for variable importance in the machine learning model.

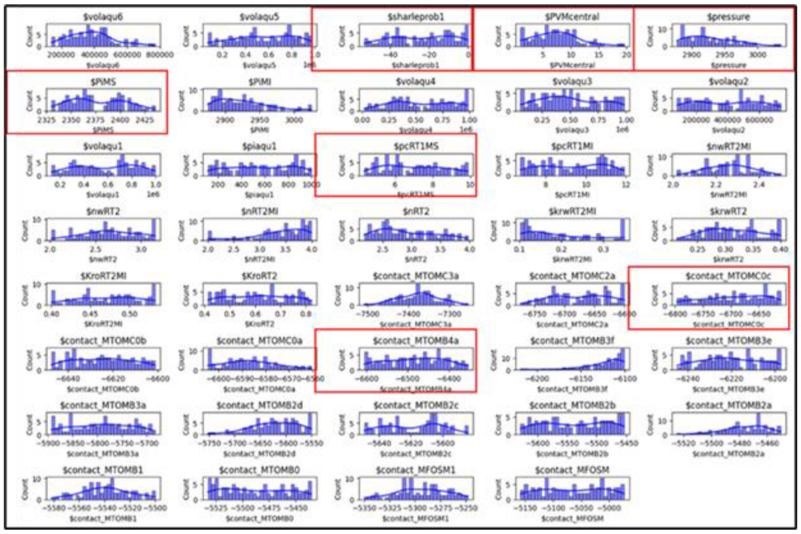


Figure 6—Posterior distribution of the optimal cases.

Fig. 7 shows the ensemble of adjusted cases after two iterations, along with the pressure match images using known data from some of the regions. These images demonstrate a clear improvement in the simulation results compared with the observed data (light blue trace).

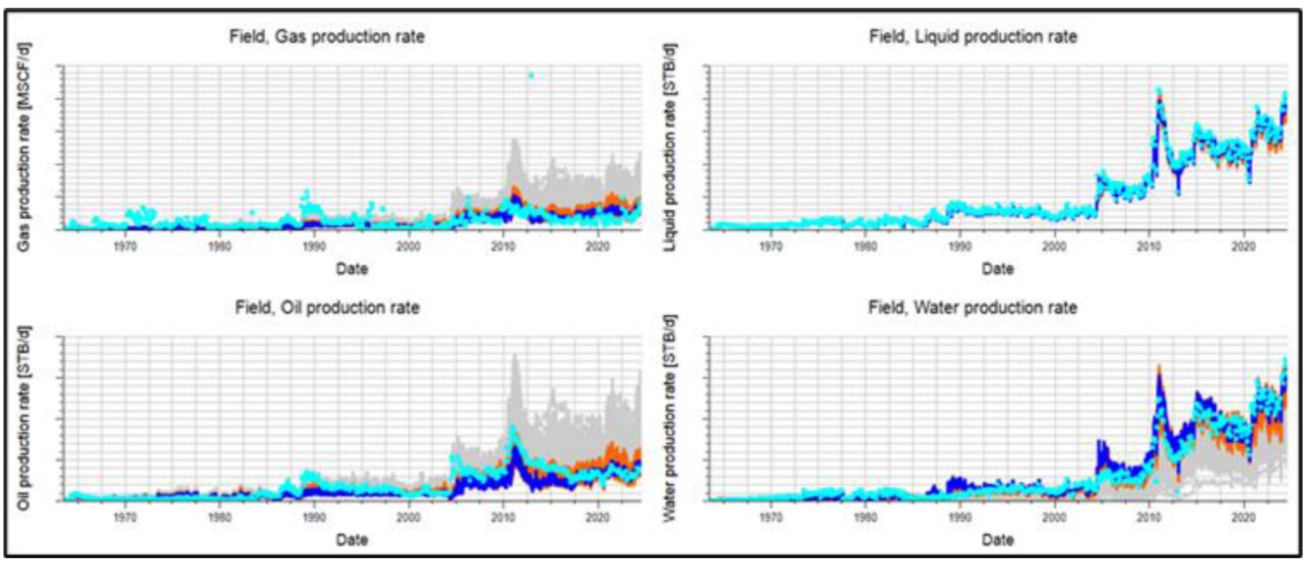


Figure 7—Ensemble of cases after two iterations (dark blue traces) compared with the first iteration (Gray traces), the second iteration (orange traces), and observed data (light blue).

Conclusions

The integration of automated workflows, the EMBER algorithm, and machine learning techniques has significantly enhanced the efficiency and accuracy of reservoir modeling and history matching in this Colombian field project. The automated workflows streamlined the process by incorporating static and dynamic parameters, reducing human error, and allowing for rapid adjustments. The use of the EMBER algorithm for property modeling provided precise estimations of porosity and permeability, improving the predictability of the model and overcoming traditional geostatistical challenges.

Machine learning played an important role in history matching by addressing the uncertainties inherent in the reservoir. The iterative process, supported by sensitivity analysis and the use of Random Forest algorithm models, enabled the identification of optimal combinations of uncertainty parameters. The SHAP analysis further refined the model by highlighting the most impactful variables, leading to more accurate and reliable simulation results.

Visual tools, such as tornado plots, worm plots, and histograms, facilitated a deeper understanding of variable significance and distribution, aiding engineers in making informed decisions. The ensemble of adjusted cases demonstrated a clear improvement in simulation results compared with observed data, validating the effectiveness of the methodologies employed.

Overall, the integrated approach not only enhanced the accuracy and efficiency of reservoir modeling but also provided a higher degree of predictability in forecast scenarios. This enhanced accuracy and efficiency is essential for quantifying risks associated with field development plans and optimizing reservoir performance. By incorporating these probabilistic scenarios into the uncertainty analysis, the study was able to present a comprehensive risk assessment, highlighting the range of possible results and their associated probabilities. This assessment not only enhanced the reliability of the predictions but also provided a solid basis for developing robust field development plans. The resulting models offered a higher degree of confidence in the forecast scenarios, ensuring that the chosen strategies were well-aligned with the overall objectives of optimizing reservoir performance and managing uncertainties effectively.

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